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Aim

To compare an original song, its karaoke (music-only) version, and a completely different song by applying ten different impulse responses using convolution, generating processed audio files, comparing their waveforms, and analyzing the effect of each impulse response.

Objectives

1. To understand the concept of convolution in Digital Signal Processing (DSP).
2. To study the effect of different impulse responses on audio signals.
3. To compare the response of three different audio signals under the same filtering conditions.
4. To generate processed audio using ten different impulse responses.
5. To compare the waveforms of the processed original, karaoke, and different songs.
6. To analyze how each impulse response modifies different types of audio signals.
7. To observe similarities and differences between the three audio signals after processing.

Theory

Convolution is a fundamental operation in Digital Signal Processing (DSP). It describes how an input signal is modified by a system represented by its impulse response.
For discrete-time signals,

$$y[n] = x[n] * h[n]$$

where
- $x[n]$= Input audio signal
- $h[n]$= Impulse response
- $y[n]$= Output audio signal

Each impulse response represents a different audio processing operation such as smoothing, filtering, reverberation, or echo. By convolving the same impulse response with different audio signals, the characteristics of each signal can be compared and analyzed.
In this experiment, three audio signals are processed:
- Original Song
- Karaoke (Music-only) Version
- Completely Different Song

Each audio signal is processed using ten different impulse responses to study their behavior under identical signal-processing conditions.

Software Requirements

- NumPy
- SciPy
- Matplotlib
- Jupyter Notebook / VS Code

Impulse Responses Used

S.N	Impulse Response	Purpose
1	Identity	Original signal without modification
2	Noise Reduction	Reduces noise by smoothing the signal
3	Low-Pass Filter	Removes high-frequency components
4	High-Pass Filter	Removes low-frequency components
5	Edge Detection	Highlights sudden signal changes
6	Sharpen	Enhances details in the waveform
7	Room Reverb	Simulates reflections in a small room
8	Hall Reverb	Simulates reflections in a large hall
9	Echo	Produces delayed repetitions of the signal
10	Exponential Decay	Produces a gradual fading effect

Algorithm

1. Import the required Python libraries.
2. Load the three WAV audio files.
3. Convert stereo audio into mono.
4. Normalize all three audio signals.
5. Make all signals equal in length.
6. Define ten different impulse responses.
7. Apply convolution separately to each audio signal using every impulse response.
8. Normalize the processed signals.
9. Save each processed signal as a WAV file.
10. Plot comparison graphs for each impulse response showing:

- Original Song
- Karaoke Song
- Different Song

11. Save all comparison graphs.
12. Analyze the effect of each impulse response.

Input

Input Audio File (.wav)

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Code and Output

```python
import os
import numpy as np
import matplotlib.pyplot as plt

from scipy.io import wavfile
from scipy.signal import convolve
```

[50] ✓ 0.0s Python

```python
song1_path = r"C:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Audio Files\abhimanyu.wav"

song2_path = r"C:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Audio Files\pv-colored.wav"

song3_path = r"C:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Audio Files\jathi_rathnalu.wav"

for file in [song1_path, song2_path, song3_path]:

    if not os.path.exists(file):
        print(file)
        raise FileNotFoundError("Audio file not found.")

print("All Audio Files Found Successfully.")

fs1, song1 = wavfile.read(song1_path)
fs2, song2 = wavfile.read(song2_path)
fs3, song3 = wavfile.read(song3_path)

if song1.ndim == 2:
    song1 = np.mean(song1, axis=1)

if song2.ndim == 2:
    song2 = np.mean(song2, axis=1)

if song3.ndim == 2:
    song3 = np.mean(song3, axis=1)

song1 = song1.astype(np.float32)
```

```python
    song2 = song2 / np.max(np.abs(song2))
    song3 = song3 / np.max(np.abs(song3))

    minimum = min(
        len(song1),
        len(song2),
        len(song3)
    )

    song1 = song1[:minimum]
    song2 = song2[:minimum]
    song3 = song3[:minimum]

    print("Sampling Rate :", fs1)
    print("Common Samples :", minimum)

    audio_folder = "Processed_Audio"
    graph_folder = "Comparison_Graphs"

    os.makedirs(audio_folder, exist_ok=True)
    os.makedirs(graph_folder, exist_ok=True)

    print("Output folders created successfully.")
```

[51] ✓ 0.0s Python

```
All Audio Files Found Successfully.
Sampling Rate : 44100
Common Samples : 1297113
Output folders created successfully.
```

```python
# 1. Identity
ir1 = np.array([1.0])

# 2. Noise Reduction (Moving Average)
ir2 = np.ones(5) / 5

# 3. Low Pass Filter
ir3 = np.array([
    0.25,
    0.50,
    0.25
])

# 4. High Pass Filter
ir4 = np.array([
    -1,
    2,
    -1
])

# 5. Edge Detection
ir5 = np.array([
    1,
    0,
    -1
])

# 6. Sharpen
ir6 = np.array([
```

```python
    -0.5,
    2.0,
    -0.5
])

# 7. Room Reverb
ir7 = np.array([
    1.0,
    0.6,
    0.3,
    0.15
])

# 8. Hall Reverb
ir8 = np.exp(
    -np.linspace(
        0,
        2,
        25
    )
)

# 9. Echo
ir9 = np.zeros(30)

ir9[0] = 1.0
ir9[10] = 0.6
ir9[20] = 0.4

# 10. Exponential Decay
ir10 = np.exp(
    -np.linspace(
        0,
        3,
        15
    )
)

impulse_responses = {

    "Identity"       : ir1,
    "NoiseReduction" : ir2,
    "LowPass"        : ir3,
    "HighPass"       : ir4,
    "EdgeDetection"  : ir5,
    "Sharpen"        : ir6,
    "RoomReverb"     : ir7,
    "HallReverb"     : ir8,
    "Echo"           : ir9,
    "Decay"          : ir10
}

print("Total Impulse Responses :", len(impulse_responses))
print("\nImpulse Responses")

for name in impulse_responses:
    print(name)
```

52] ✓ 0.0s Python

[52] ✓ 0.0s Python

```
Total Impulse Responses : 10

Impulse Responses
Identity
NoiseReduction
LowPass
HighPass
EdgeDetection
Sharpen
RoomReverb
HallReverb
Echo
Decay
```

```python
original_outputs = {}

for name, ir in impulse_responses.items():

    output = convolve(song1, ir, mode="same")

    output = output / np.max(np.abs(output))

    original_outputs[name] = output

    wavfile.write(
        os.path.join(
            audio_folder,
            f"Original_{name}.wav"
        ),
        fs1,
        (output * 32767).astype(np.int16)
    )

print("Original Song Processing Completed")

print("Generated Files")

for name in impulse_responses.keys():
    print(f"Original_{name}.wav")
```

[53] ✓ 0.2s Python

```
Original Song Processing Completed
Generated Files
Original_Identity.wav
Original_NoiseReduction.wav
Original_LowPass.wav
Original_HighPass.wav
Original_EdgeDetection.wav
Original_Sharpen.wav
Original_RoomReverb.wav
Original_HallReverb.wav
Original_Echo.wav
Original_Decay.wav
```

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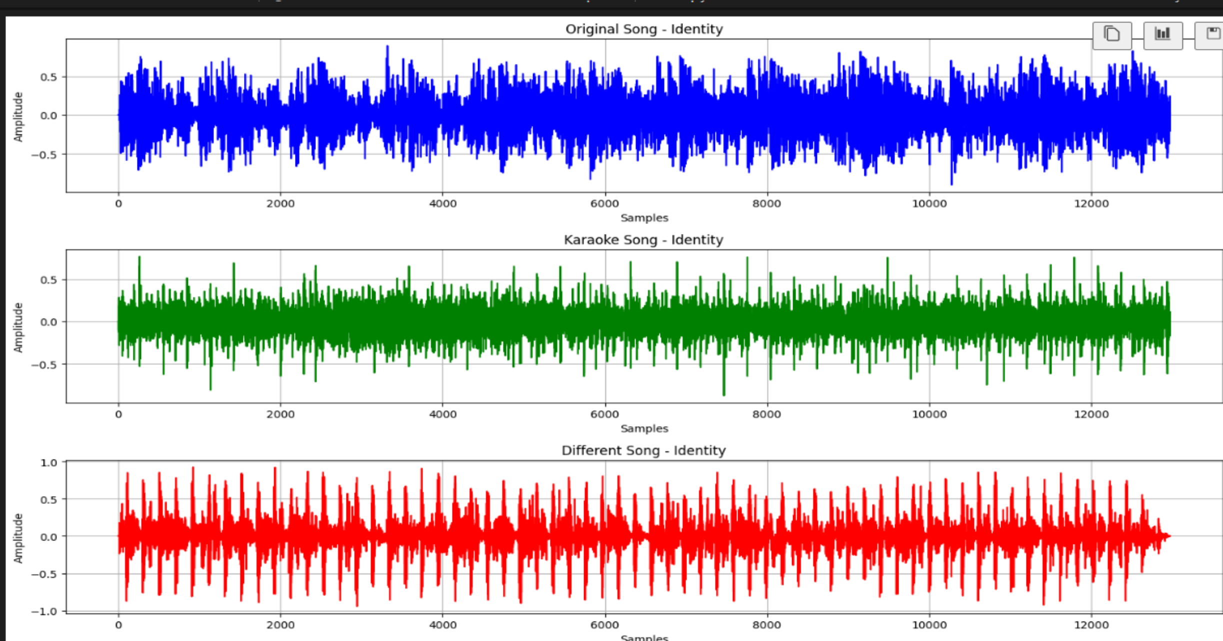
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["Identity"][::100], color="blue")
plt.title("Original Song - Identity")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["Identity"][::100], color="green")
plt.title("Karaoke Song - Identity")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["Identity"][::100], color="red")
plt.title("Different Song - Identity")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "Identity_comparison.png"
    )
)

plt.show()

print("Identity Comparison Saved")
```

✓ 0.7s Python



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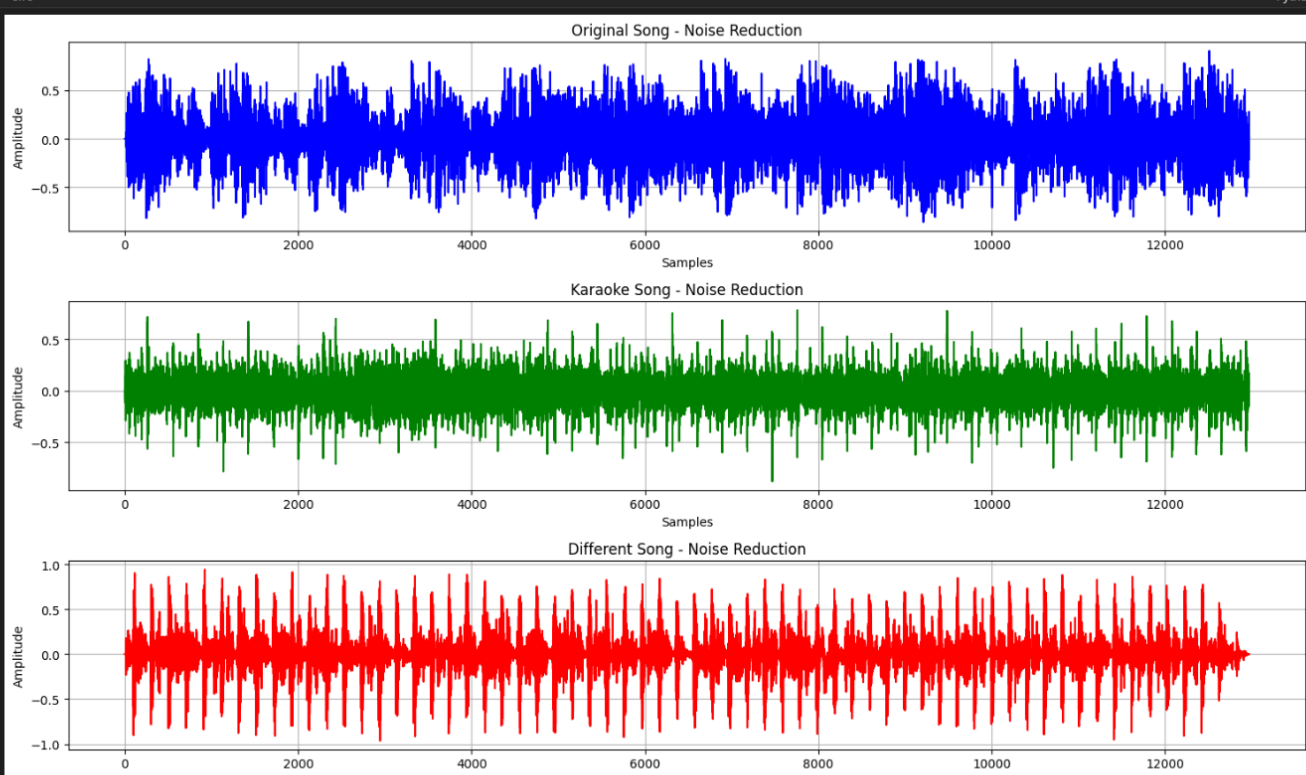
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["NoiseReduction"][::100], color="blue")
plt.title("Original Song - Noise Reduction")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["NoiseReduction"][::100], color="green")
plt.title("Karaoke Song - Noise Reduction")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["NoiseReduction"][::100], color="red")
plt.title("Different Song - Noise Reduction")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "NoiseReduction_Comparison.png"
    )
)

plt.show()

print("Noise Reduction Comparison Saved")
```

✓ 0.7s Python



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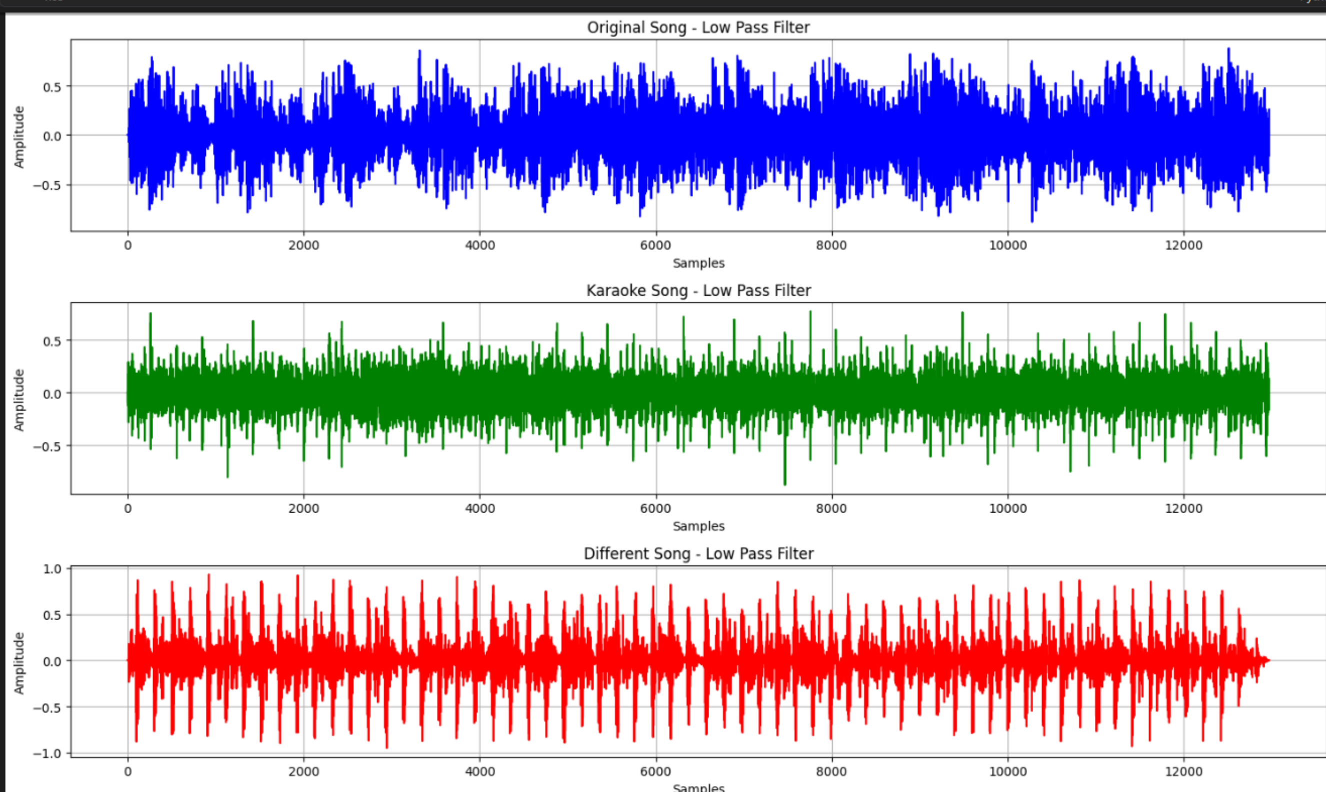
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["LowPass"][::100], color="blue")
plt.title("Original Song - Low Pass Filter")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["LowPass"][::100], color="green")
plt.title("Karaoke Song - Low Pass Filter")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["LowPass"][::100], color="red")
plt.title("Different Song - Low Pass Filter")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "LowPass_Comparison.png"
    )
)

plt.show()

print("Low Pass Comparison Saved")
```

[58] ✓ 1.0s Python



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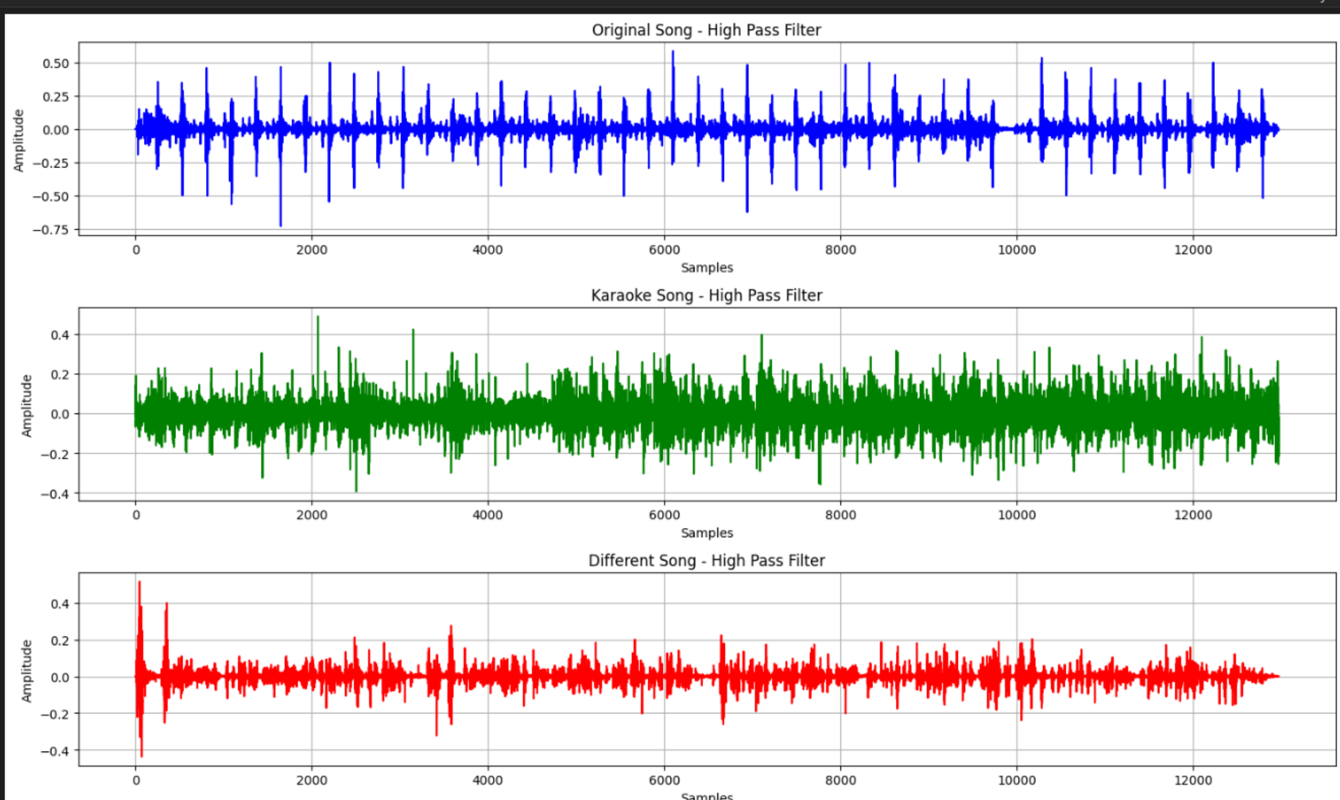
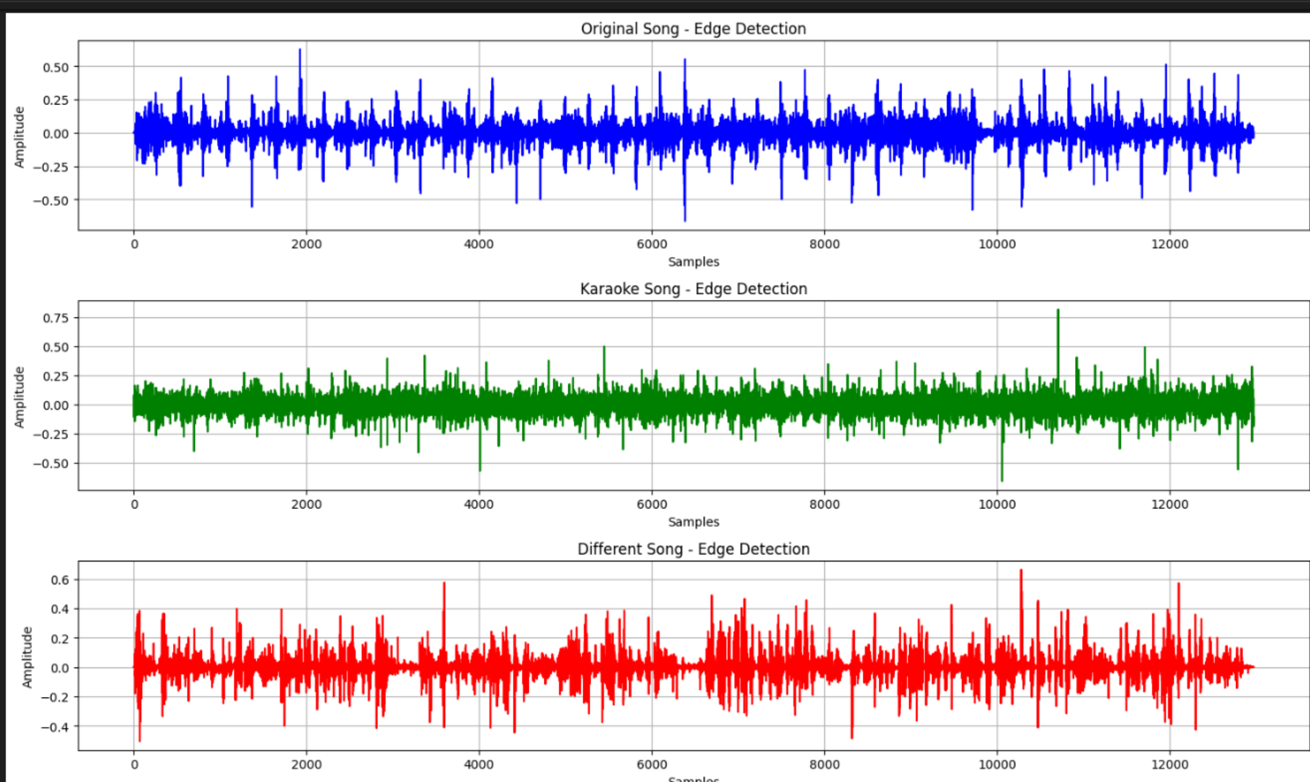
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["HighPass"][::100], color="blue")
plt.title("Original Song - High Pass Filter")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["HighPass"][::100], color="green")
plt.title("Karaoke Song - High Pass Filter")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["HighPass"][::100], color="red")
plt.title("Different Song - High Pass Filter")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "HighPass_Comparison.png"
    )
)

plt.show()

print("High Pass Comparison Saved")
```

[59] ✓ 0.5s Python



```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["EdgeDetection"][::100], color="blue")
plt.title("Original Song - Edge Detection")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["EdgeDetection"][::100], color="green")
plt.title("Karaoke Song - Edge Detection")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["EdgeDetection"][::100], color="red")
plt.title("Different Song - Edge Detection")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "EdgeDetection_Comparison.png"
    )
)

plt.show()

print("Edge Detection Comparison Saved")
```

[60] ✓ 1.3s Python



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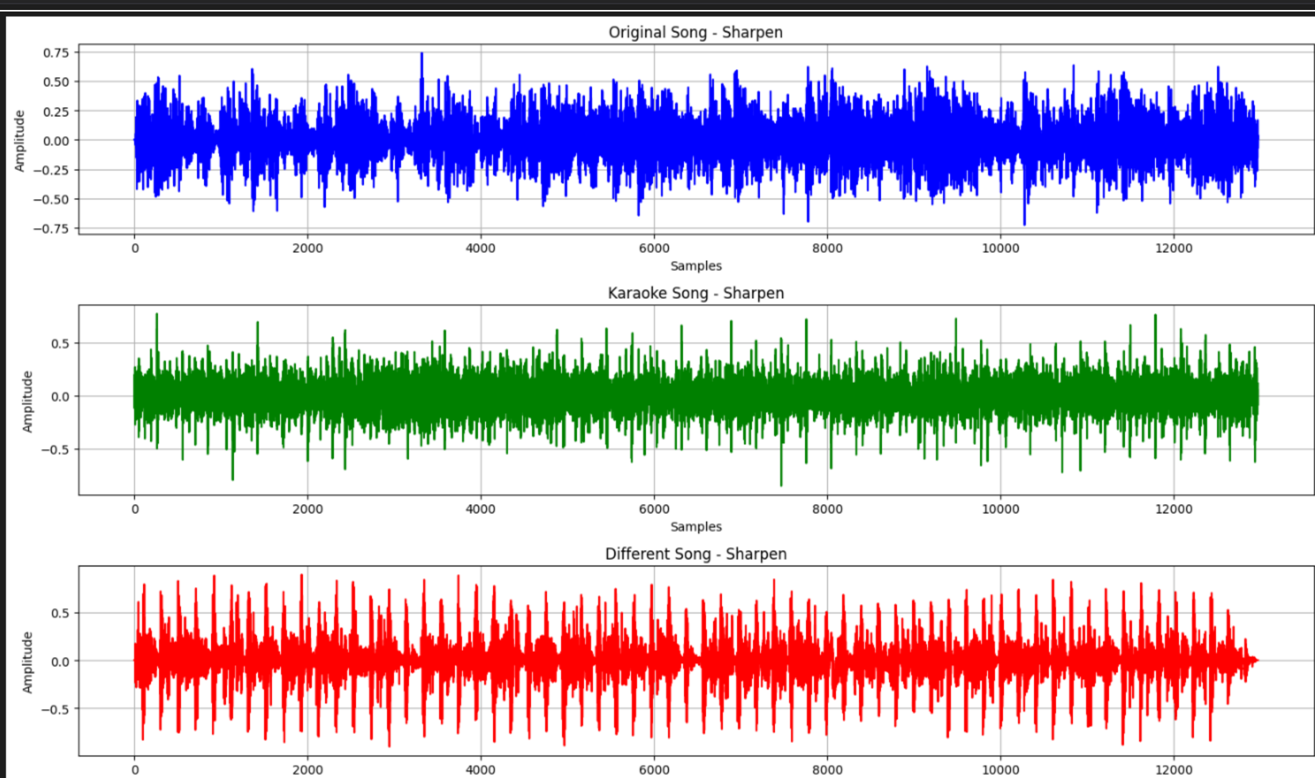
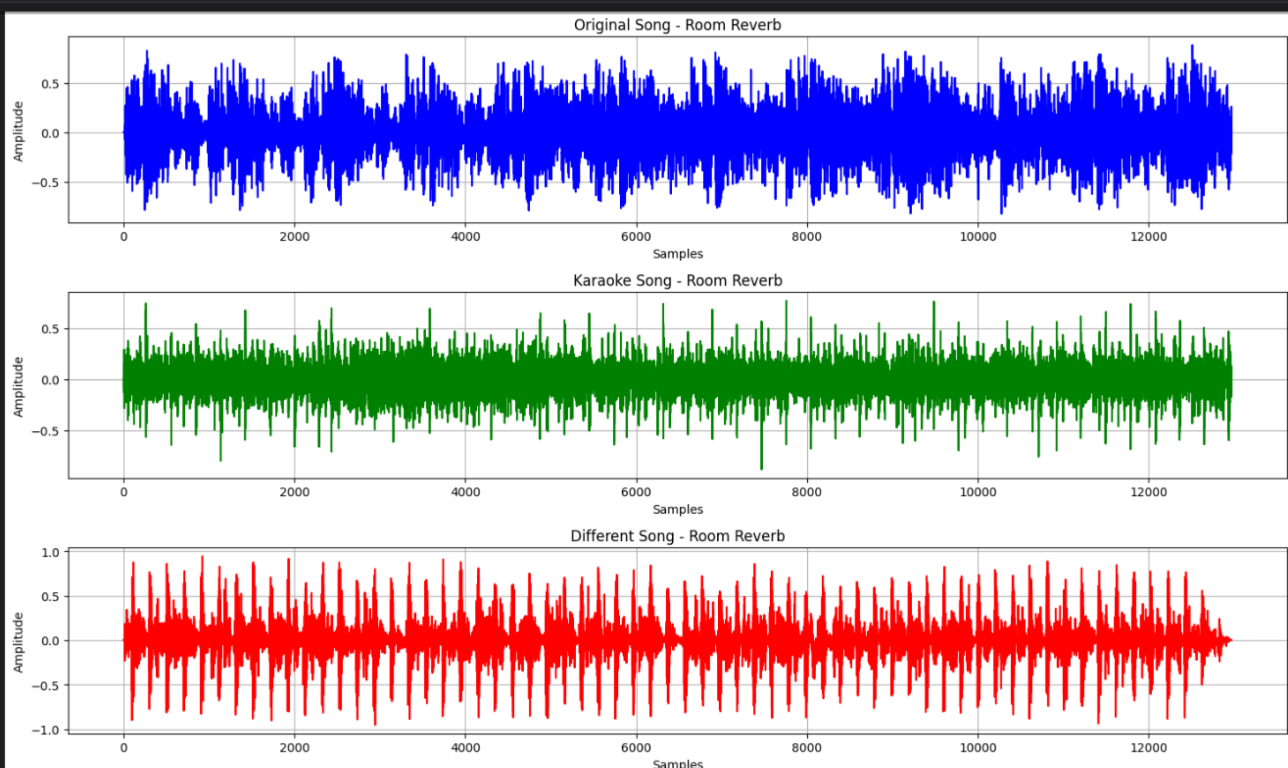
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["Sharpen"][::100], color="blue")
plt.title("Original Song - Sharpen")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["Sharpen"][::100], color="green")
plt.title("Karaoke Song - Sharpen")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["Sharpen"][::100], color="red")
plt.title("Different Song - Sharpen")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "Sharpen_Comparison.png"
    )
)

plt.show()

print("Sharpen Comparison Saved")
```

[61] ✓ 0.6s Python



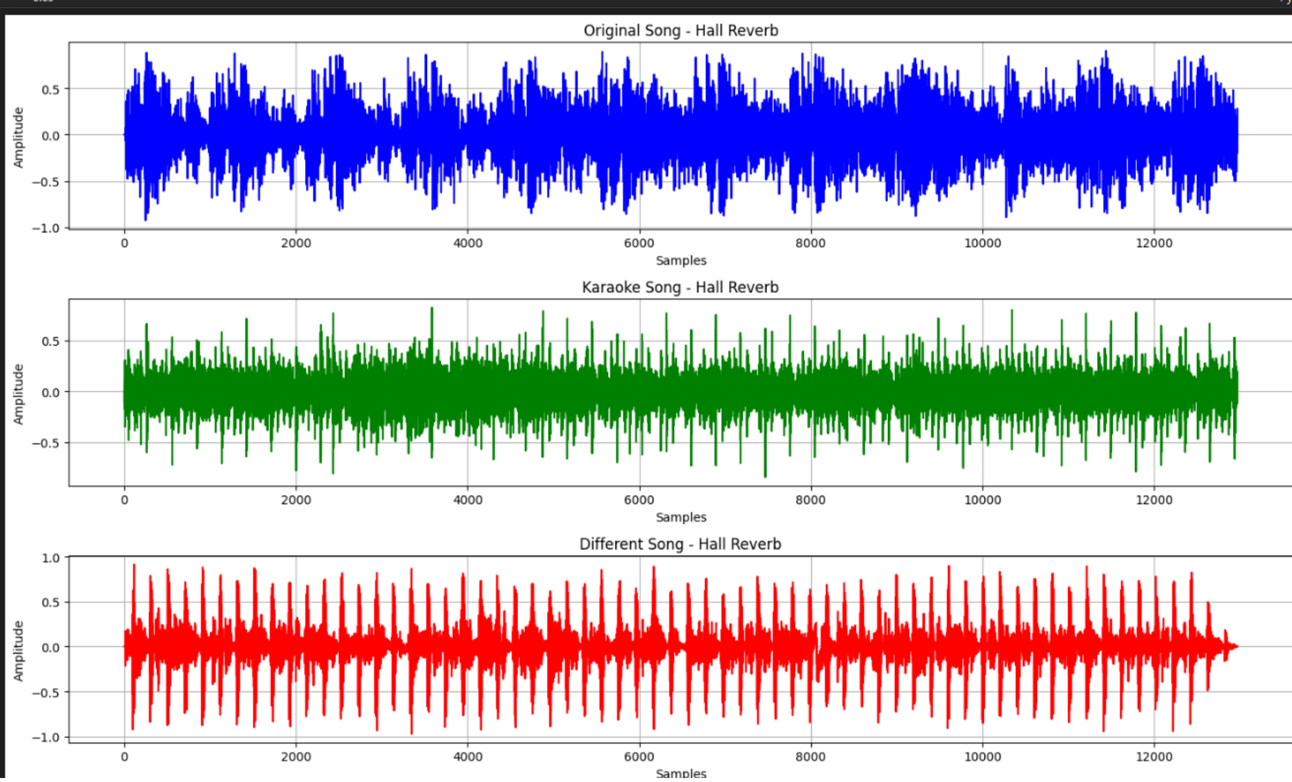
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["RoomReverb"][::100], color="blue")
plt.title("Original Song - Room Reverb")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["RoomReverb"][::100], color="green")
plt.title("Karaoke Song - Room Reverb")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["RoomReverb"][::100], color="red")
plt.title("Different Song - Room Reverb")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "RoomReverb_Comparison.png"
    )
)

plt.show()

print("Room Reverb Comparison Saved")
```

[62] ✓ 0.6s Python



```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["HallReverb"][::100], color="blue")
plt.title("Original Song - Hall Reverb")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["HallReverb"][::100], color="green")
plt.title("Karaoke Song - Hall Reverb")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["HallReverb"][::100], color="red")
plt.title("Different Song - Hall Reverb")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "HallReverb_Comparison.png"
    )
)

plt.show()

print("Hall Reverb Comparison Saved")
```

[63] ✓ 0.6s Python



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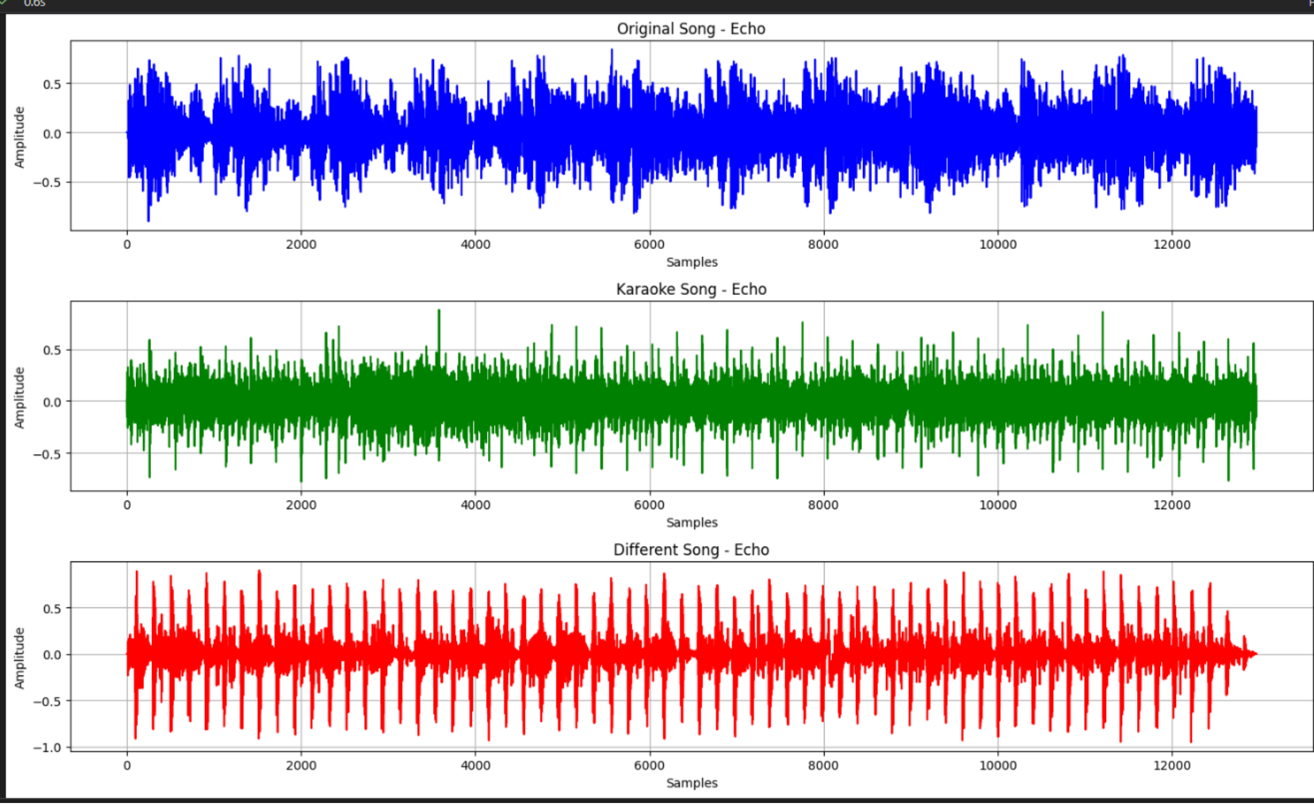
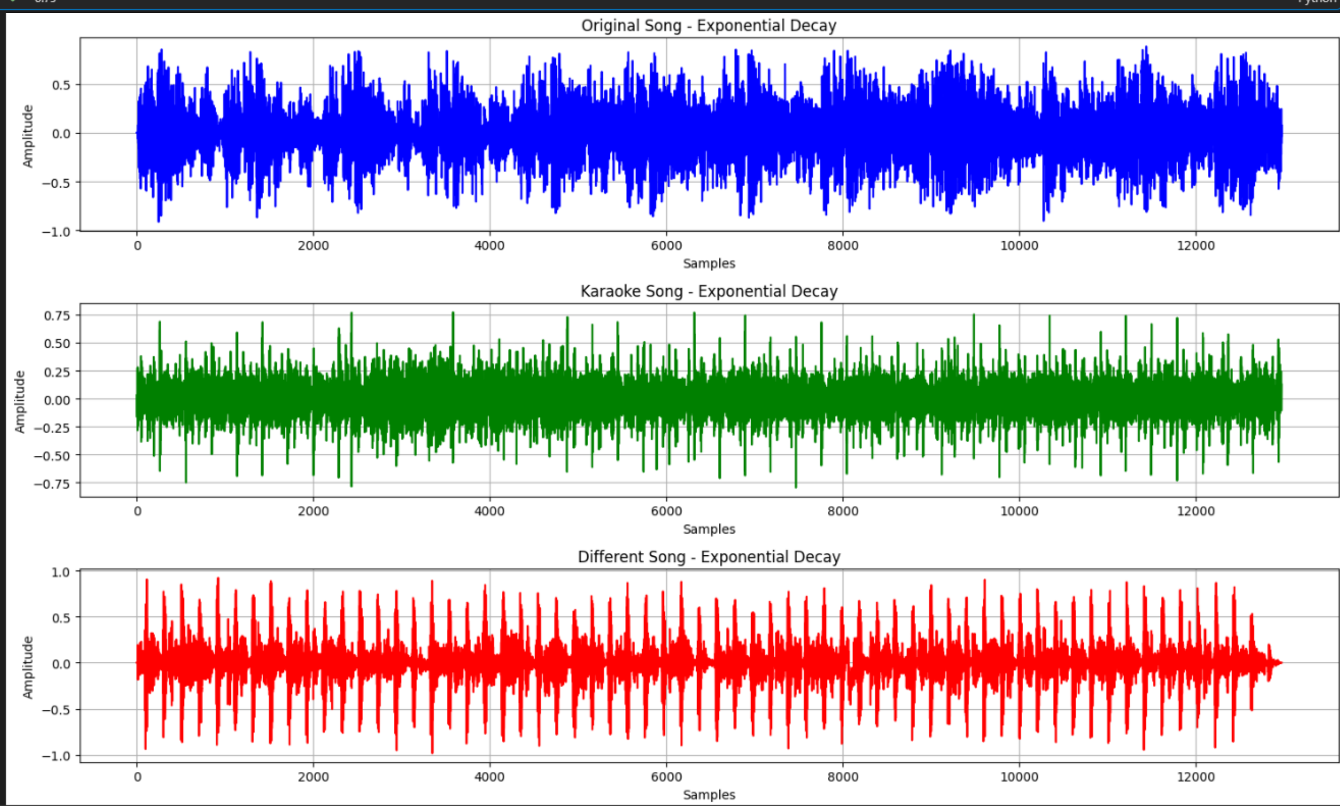
```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["Echo"][::100], color="blue")
plt.title("Original Song - Echo")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["Echo"][::100], color="green")
plt.title("Karaoke Song - Echo")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["Echo"][::100], color="red")
plt.title("Different Song - Echo")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "Echo_Comparison.png"
    )
)

plt.show()

print("Echo Comparison Saved")
```

[64] ✓ 0.6s Python



```python
plt.figure(figsize=(15,9))

plt.subplot(3,1,1)
plt.plot(original_outputs["Decay"][::100], color="blue")
plt.title("Original Song - Exponential Decay")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(karaoke_outputs["Decay"][::100], color="green")
plt.title("Karaoke Song - Exponential Decay")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(different_outputs["Decay"][::100], color="red")
plt.title("Different Song - Exponential Decay")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid(True)

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "Decay_Comparison.png"
    )
)

plt.show()

print("Exponential Decay Comparison Saved")
```

[65] ✓ 0.7s Python



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```python
print("PROCESS COMPLETED SUCCESSFULLY")

print("\nGenerated Audio Files")

for name in impulse_responses.keys():

    print(f"Original_{name}.wav")
    print(f"Karaoke_{name}.wav")
    print(f"Different_{name}.wav")

print("\nGenerated Comparison Images")

print("Identity_Comparison.png")
print("NoiseReduction_Comparison.png")
print("LowPass_Comparison.png")
print("HighPass_Comparison.png")
print("EdgeDetection_Comparison.png")
print("Sharpen_Comparison.png")
print("RoomReverb_Comparison.png")
print("HallReverb_Comparison.png")
print("Echo_Comparison.png")
print("Decay_Comparison.png")

print("\nTotal Audio Files :", 30)
print("Total Comparison Images :", 10)

print("\nAudio Folder :", os.path.abspath(audio_folder))
print("Graph Folder :", os.path.abspath(graph_folder))
```

[67] ✓ 0.0s Python

```
PROCESS COMPLETED SUCCESSFULLY

Generated Audio Files
Original_Identity.wav
Karaoke_Identity.wav
Different_Identity.wav
Original_NoiseReduction.wav
Karaoke_NoiseReduction.wav
Different_NoiseReduction.wav
Original_LowPass.wav
Karaoke_LowPass.wav
Different_LowPass.wav
Original_HighPass.wav
Karaoke_HighPass.wav
Different_HighPass.wav
Original_EdgeDetection.wav
Karaoke_EdgeDetection.wav
Different_EdgeDetection.wav
Original_Sharpen.wav
Karaoke_Sharpen.wav
Different_Sharpen.wav
Original_RoomReverb.wav
Karaoke_RoomReverb.wav
Different_RoomReverb.wav
Original_HallReverb.wav
...
Total Comparison Images : 10

Audio Folder : c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Processed_Audio
Graph Folder : c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Comparison_Graphs
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Output

The program generates:

- 30 processed audio files
- 10 waveform comparison images
- Processed audio for each impulse response

Observations

Identity

The processed signals remain almost identical to their respective original audio.

Noise Reduction

All three audio signals become smoother with reduced noise and small fluctuations.

Low-Pass Filter

High-frequency components are attenuated, producing smoother waveforms and softer audio.

High-Pass Filter

Low-frequency components are reduced, emphasizing sharper transitions and high-frequency details.

Edge Detection

Rapid changes in the waveform become more prominent, making transients easier to identify.

Sharpen

Enhances waveform details and improves the clarity of audio transitions.

Room Reverb

Introduces mild reverberation, simulating the acoustic effect of a small room.

Hall Reverb

Produces stronger reverberation with longer reflections similar to a large hall.

Echo

Creates delayed repetitions of the original sound, increasing the apparent depth of the audio.

Exponential Decay

Applies a gradually decreasing response, resulting in a fading reverberation effect.

Comparison Table

Impulse Response	Original Song	Karaoke Song	Different Song
Identity	No noticeable change	No noticeable change	No noticeable change
Noise Reduction	Smoother waveform	Music becomes smoother	Background noise reduced
Low Pass	Treble reduced	High-frequency music reduced	Smoother output
High Pass	Bass reduced	Percussion emphasized	High-frequency details enhanced
Edge Detection	Speech transitions highlighted	Instrument transitions highlighted	Sudden waveform changes enhanced
Sharpen	Improved clarity	Instrument details enhanced	Waveform details increased
Room Reverb	Mild reverberation	Natural room effect	Slight echo effect
Hall Reverb	Long reverberation	Large hall ambience	Extended reflections
Echo	Audible delayed repetition	Echo in instruments	Echo effect visible
Exponential Decay	Gradual fading	Smooth decay of music	Progressive amplitude reduction

Result

The three audio signals were successfully processed using ten different impulse responses. Thirty processed audio files and ten waveform comparison graphs were generated. The comparison demonstrated that each impulse response affects the original song, karaoke version, and different song differently depending on their frequency content and temporal characteristics.

Conclusion

The experiment successfully demonstrated the application of convolution using ten different impulse responses on three different audio signals. The generated comparison graphs and processed audio files showed that each impulse response produces a distinct effect on the waveform and sound quality. Filters such as Low-Pass and Noise Reduction smooth the signals, High-Pass and Sharpen emphasize finer details, while Room Reverb, Hall Reverb, Echo, and Exponential Decay introduce realistic acoustic effects. Comparing the original song, karaoke version, and a completely different song under identical processing conditions provided a clear understanding of how convolution-based filtering influences different types of audio signals and highlighted the importance of impulse responses in practical digital audio processing.

GitHub:

https://github.com/Devaharshagoud/DSIP-Experiments.git